Project Report : January 2025

Submitted to: AidKriya Foundation

Submitted by:
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Title of the Project:
AidKriya Donation Management System: A Time-Series Modeling Approach

Introduction to the Problem

The AidKRIYA Foundation's mission to uplift lives through philanthropy is truly inspiring, reminding us all of the power of giving. However, managing donations through manual processes can make things harder than they need to be. It becomes challenging to truly understand donor behavior or reach out to the right people at the right time. These hurdles can make it difficult for the foundation to maintain the steady flow of support needed to drive their life-changing work. By finding ways to simplify and improve these processes, we can help them connect with donors more effectively and make an even greater impact on the lives they touch.

Proposed Solution

Motivated by a profound empathy for AidKRIYA's mission, we embarked on a journey to harness the transformative potential of AI to address the challenges in their donation management processes. Recognizing the critical role that timely and strategic donor engagement plays in sustaining their humanitarian efforts, we aimed to create a solution that would empower the foundation to achieve greater efficiency and impact.

Our solution centers around an AI-powered system designed to predict donation patterns and recommend optimal times to engage with donors. By leveraging historical donation data, analyzing donor sentiments, and factoring in the influence of cultural and regional events, the system provides actionable insights to guide targeted outreach efforts.

This innovation not only streamlines donation management but also enables the foundation to build deeper, more meaningful connections with their supporters. By optimizing their fundraising campaigns, AidKRIYA can ensure a consistent flow of critical resources, further strengthening their ability to uplift lives and fulfill their mission of making a difference.

Steps Taken So Far

To ensure the success of our solution, we conducted research across multiple platforms, gathering a dataset that provided deep insights into the dynamics of donor behavior. This dataset was carefully curated to include a wide range of variables, such as historical donation patterns, donor sentiment analysis, contribution amounts, and context-specific factors like cultural and regional events.

We designed and implemented a SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous Variables) model. This advanced statistical model was chosen for its ability to handle time-series data with seasonal trends and external influencing factors. By incorporating these diverse variables, the SARIMAX model allowed us to analyze and forecast donation peaks with remarkable accuracy and reliability.

In addition to building the predictive model, we developed a user interface using Tkinter to enhance accessibility and usability. This UI allows users to input a prediction date and receive a forecasted donation amount for that specific date. The model is capable of predicting donations up to a range of one year, and its predictive range expands as new data is added and updated by the user. This feature ensures continuous improvement in prediction accuracy while making it easy for AidKRIYA to plan their fundraising efforts.

The package can be used by the users by following few steps. After downloading the folder, the user have to open the ‘dist’ folder. There will be a package named ‘aidKriya prediction model’. By clicking on this package, a window opens which is the User interface of the prediction model. The user can enter the prediction date range up-to one year. The forecasted value will be displayed below. The prediction range increases as the user adds more data to the dataset inside the same folder.

This approach not only provided precise predictions but also offered valuable insights into the underlying factors driving donor behavior. These insights empower AidKRIYA to plan targeted outreach campaigns, optimize resource allocation, and strengthen their overall fundraising strategy, ensuring a steady and sustainable flow of support for their humanitarian initiatives.

Future Vision and Usability

Our priority is to ensure the model is not just robust but also user-friendly. We aim to create a solution that AidKRIYA can seamlessly integrate into their operations, empowering them to unlock the full potential of their donor network. By converting the model into an intuitive tool—accessible via a simple executable file with clear instructions—we ensure that even non-technical members of the team can leverage the insights effectively.

Concluding Remarks

This project is a testament to our collective belief in using technology to drive impactful change. By integrating an intuitive UI and an adaptive predictive model, we have enhanced AidKRIYA’s ability to plan and manage their donation campaigns efficiently. Together with AidKRIYA, we hope to create a future where every effort

counts, and every donation transforms lives.

Photograph of the Team